

Take-Home Assignment

Junior AI Engineer — Build “SiGMA Event Concierge”

Duration: 2 days · Expected effort: 8–12 hours · Confidential – for candidate use only

1 · The brief

SiGMA runs the world's largest iGaming events — thousands of delegates, hundreds of sessions, speakers and exhibitors per summit. Your task is to build the **SiGMA Event Concierge**: a simple, polished web UI where an attendee can ask natural-language questions about an event and get accurate, AI-generated answers grounded in the event data we provide.

Example questions your assistant should handle: *“What AI-related talks are on Wednesday afternoon?”*, *“Who is speaking about regulation?”*, *“I’m a payments startup — which exhibitors should I visit?”*, *“Build me a 1-day agenda focused on emerging tech.”*

2 · What we provide

- **sigma_agenda.json** — a synthetic dataset for a fictional “SiGMA Malta 2026” summit: ~40 conference sessions (title, track, day, time, room, speakers, abstract) and ~20 exhibitors (name, category, stand, description).
- This brief, including the evaluation rubric below — we mark against it, nothing hidden.

3 · Core requirements

Product

- A web UI with a chat-style or ask-a-question interaction. Any stack you like — React/Vue, plain HTML/JS, or Streamlit/Gradio are all acceptable. We care about clarity and robustness, not pixel-perfection.
- Answers must be **grounded in the dataset**: the assistant should not invent sessions, speakers or exhibitors, and should say so when the data doesn't contain an answer.
- Sensible UX states: loading/streaming indicator, empty state, and graceful error handling when the model or backend fails.

Engineering

- An LLM must be involved — how you get relevant data to it (full-context stuffing, retrieval/embeddings, tool/function calling) is your design decision. Tell us why you chose your approach.
- **No paid API required.** Use a local model (e.g. Ollama), a free-tier hosted API, or both. Structure the code so the model provider is swappable (a thin adapter/interface is enough).
- A README covering: setup & run instructions, your architecture and AI-design decisions, known limitations, and what you would do next with more time.

4 · Stretch goals — pick at most two

These are optional. A small, solid core beats a sprawling half-finished one.

- **Citations**: each answer links to the session(s)/exhibitor(s) it was based on.
- **Caching**: avoid redundant model calls for repeated or near-duplicate questions.
- **Mini eval**: a small script with 8–10 test questions and a way to check groundedness/accuracy.
- **Streaming**: token-by-token response streaming to the UI.
- **Personal agenda**: let the user pin recommended sessions into a day plan.

5 · Rules of engagement

- **Timebox:** you have 2 calendar days from receiving this brief; we expect roughly 8–12 hours of work. Please don't gold-plate — scoping is part of the test.
- **AI coding assistants (Copilot, Claude, ChatGPT, etc.) are allowed and encouraged** — we use them too. You must understand every line you submit: the follow-up interview is a deep walkthrough of your code and decisions.
- **Out of scope:** authentication, deployment/hosting, databases beyond the provided JSON, and mobile apps. Don't build them.
- **Submission:** a link to a Git repository (public or invite us) with clear commit history, plus an optional ≤ 3 -minute screen recording of the app working. If you used a local model, the README must let us reproduce it, or include a mock/recorded mode.

6 · How we evaluate

Dimension	What we look for	Weight
AI integration & grounding	Relevant data reaches the model; answers are accurate and hallucination-resistant; prompt/retrieval design is deliberate and explained.	30%
Engineering quality	Clean structure, readable code, sensible abstractions (e.g. swappable model provider), error handling, meaningful commits.	25%
Product & UX judgment	The UI is usable and honest about uncertainty; loading/error/empty states; the assistant is actually helpful for an attendee.	15%
Scoping & prioritisation	A finished core within the timebox; smart trade-offs; stretch goals only where the core is solid.	15%
Communication	README quality; clarity of design decisions and limitations; the code walkthrough in the follow-up interview.	15%

7 · Timeline & next steps

- **Day 0:** you receive this brief and the dataset; confirm receipt by email.
- **Day 2 (by 18:00 CET):** submit your repository link (and optional demo video).
- **Within 3 working days:** a 45-minute follow-up call — live demo, code walkthrough, and discussion of what you'd change for production.

Questions during the task are welcome — asking good questions counts in your favour, not against you. Contact: **Antoine Borg**, antoine.b@sigma.world.

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